Probability and Statistics - Definitions

Dr. D Bhanu Prakash

The Data Detective: A Journey Through Probability

Framing device: The class becomes a junior detective agency for one continuous case. Every concept is a new "instrument" unlocked to crack the file.

The Case: "Last night someone tampered with the vending machine's lock. Campus security has narrowed it to 4 suspects: Alex, Bo, Chris, and Dana. There's a muddy shoe print, a security camera that is only half-recorded, a fingerprint smudge, an alibi to check, and an anonymous tip. By the end of this lecture you'll have the exact toolkit to find out who did it."

PART A — THE CASE

0. The Hook

"While waiting to be questioned, Dana pulls out a coin: 'Let's flip for who talks first.' It comes up heads 10 times in a row. Someone in the room mutters, *'that coin's rigged.'* Is it?"

---

1. Sample Space & Events

Definition — Sample space (S): the set of *all* possible outcomes of an experiment or process. **Definition — Event:** any subset of S; a collection of outcomes you actually care about.

Case application: The suspect pool is S = {Alex, Bo, Chris, Dana}. The event "has muddy shoes" is A = {Alex, Chris}. The event "was seen near the machine" is B = {Chris, Dana}.

Introduce set notation with a quick Venn sketch: **union** ($A \cup B$, "or"), **intersection** ($A \cap B$, "and"), **complement** (A^c, "not A"). Case check: $A \cap B$ = {Chris} — the overlap is already narrowing the case.

---

2. Counting Techniques

Definition — Multiplication principle: if a process has k steps with $n_1, n_2, \ldots, n_k$ choices at each step, the total number of outcomes is $n_1 \times n_2 \times \ldots \times n_k$. **Definition — Permutation** $P(n,r) = n! / (n - r)!$: number of ordered arrangements of r items chosen from n. **Definition — Combination** $C(n,r) = n! / [r!(n-r)!]$: number of unordered selections of r items from n.

Case application: - Security has 4 suspects and wants to rank all of them from most to least suspicious → $P(4,4) = 4! = 24$ possible rankings. - Security can only interview 2 of the 4 suspects today → $C(4,2) = 6$ possible interview pairs. List them on the board: {A,B},{A,C},{A,D},{B,C},{B,D},{C,D}. - The evidence locker has 3 shelves and each holds one of 5 evidence bags in some order → $5 \times 4 \times 3 = 60$ arrangements (permutation shortcut).

Point out explicitly: *"Permutations care about order — who's ranked #1 suspect. Combinations don't — which pair gets interviewed, regardless of who's asked first."*

---

3. Axioms of Probability

Axiom 1: $P(A) \geq 0$ for every event A. **Axiom 2:** $P(S) = 1$. **Axiom 3:** If A and B are mutually exclusive ($A \cap B$ = the empty set, i.e. they share no outcomes), then $P(A \cup B) = P(A) + P(B)$.

Case application: Detective assigns initial suspicion P(Alex)=0.25, P(Bo)=0.25, P(Chris)=0.25, P(Dana)=0.25 — exactly one of them did it, so these must sum to 1 (Axiom 2), and no suspicion can be negative (Axiom 1). If new evidence made someone's number negative or the total exceeded 1, the detective's math would be *wrong*, not just surprising.

---

4. Addition Rules

General addition rule: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$. **Special case (mutually exclusive):** $P(A \cup B) = P(A) + P(B)$.

Case application: P(muddy shoes) = 0.5, P(near machine) = 0.5, P(both) = 0.25.

P(muddy OR near machine) = 0.5 + 0.5 − 0.25 = **0.75**

"Notice we don't just add — Chris satisfies *both* conditions, and if we didn't subtract the overlap, we'd count Chris's suspicion twice."

---

5. Multiplication Rules & Conditional Probability

This is the heart of real detective work: *how does one clue change the odds on another?* Several ideas build on each other here — take these one at a time.

Definition — Conditional probability: $P(A \mid B) = P(A \cap B) / P(B)$, for $P(B) > 0$. Read "the probability of A, given that B has occurred." Conditioning *shrinks* the sample space down to just the world where B is true.

Case application 1: Of the 2 suspects near the machine (Chris, Dana), only Chris also has muddy shoes. P(muddy | near machine) = P(muddy ∩ near) / P(near) = 0.25 / 0.5 = **0.5**

Compare to the plain P(muddy shoes) = 0.5 — here it happens to be unchanged, but flag that this usually *does* shift, which is exactly why conditioning matters.

> **General Multiplication Rule (two events):** $P(A \cap B) = P(A) \cdot P(B \mid A) = P(B) \cdot P(A \mid B)$. This is just the conditional probability definition rearranged — useful when $P(B|A)$ is easier to know directly than $P(A \cap B)$.
>
> **Chain Rule (multiplication rule for several events):** for events $A_1, A_2, \ldots, A_n$, $P(A_1 \cap A_2 \cap \ldots \cap A_n) = P(A_1) \cdot P(A_2 \mid A_1) \cdot P(A_3 \mid A_1 \cap A_2) \ldots P(A_n \mid A_1 \cap \ldots \cap A_{n-1})$. Each new clue's probability is conditioned on *everything found before it.*

Case application 2 — three sequential clues on suspect Chris: - P(fingerprint matches) = 0.4 - P(alibi fails | fingerprint matches) = 0.6 - P(camera partially shows Chris | fingerprint matches AND alibi fails) = 0.5

P(all three clues point to Chris) = 0.4 × 0.6 × 0.5 = **0.12**

Draw this live as a **tree diagram**: first branch (fingerprint yes/no), second branch off each (alibi fails/holds), third branch off each of those (camera shows/doesn't). "Every path down the tree is a chain-rule multiplication — this is how detectives (and data scientists) combine sequential evidence."

> **Sampling with vs. without replacement:** if the detective draws 2 pieces of evidence at random from a bag of 5 (3 relevant, 2 irrelevant) *without* replacement, the second draw's probabilities depend on the first — conditioning is unavoidable. *With* replacement, the draws don't affect each other. This distinction is the seed of the independence definition coming next.
>
> **Law of Total Probability (bridge to Bayes):** if $B_1, B_2, \ldots, B_n$ partition the sample space (mutually exclusive, covering everything), then $P(A) = P(A \mid B_1)P(B_1) + P(A \mid B_2)P(B_2) + \ldots + P(A \mid B_n)P(B_n)$. **Case framing:** "The tip could name Dana for two disjoint reasons: she's guilty and correctly named, or she's innocent and wrongly named. Adding those two paths gives the *total* probability the tip names her at all." (We'll use this exact idea in Section 7.)

---

6. Independence

> **Definition:** Events A and B are independent iff $P(A \mid B) = P(A)$, equivalently $P(A \cap B) = P(A) \cdot P(B)$. Learning that B happened tells you *nothing new* about A.

Case application — back to the coin: Each flip of Dana's coin is independent of the last. Getting 10 heads in a row from a fair coin has probability $(1/2)^{10} = 1/1024$ — rare, but not impossible, and it says *nothing* about flip #11, which is still 50/50. This is the classic **gambler's fallacy**: thinking "it's due" after a streak.

Quick contrast: is "muddy shoes" independent of "near the machine" in our case? Check: P(muddy) × P(near) = 0.5 × 0.5 = 0.25, and we already found P(muddy ∩ near) = 0.25. **They match — so, for this case's numbers, these two clues are independent**, which is itself useful detective information: muddy shoes don't make "near the machine" any more or less likely.

---

7. Bayes' Formula — *the climax*

The tip: An anonymous tipster — right 90% of the time — names Dana. Gut reaction: "so Dana is 90% likely guilty." That instinct is almost always wrong, and Bayes' formula shows exactly why.

> **Bayes' Formula:** $P(A \mid B) = [P(B \mid A) \cdot P(A)] / P(B)$, where $P(B)$ is expanded using the Law of Total Probability.

Setting up the case numbers: - Prior: $P(G) = P(\text{Dana guilty}) = 1/4 = 0.25 \rightarrow P(G') = 0.75$ (before any tip, any one of the 4 is equally likely) - $P(T \mid G) = 0.90$ — if Dana is guilty, the tipster correctly names her 90% of the time - $P(T \mid G') = 0.10$ — if Dana is innocent, the tipster still (wrongly) names her 10% of the time

Step 1 — find P(T) via the Law of Total Probability: $P(T) = P(T|G)\cdot P(G) + P(T|G')\cdot P(G') = (0.90)(0.25) + (0.10)(0.75) = 0.225 + 0.075 = \mathbf{0.30}$

Step 2 — apply Bayes' formula: $P(G \mid T) = P(T|G)\cdot P(G) / P(T) = 0.225 / 0.30 = \mathbf{0.75}$

The punchline: even with a "90% accurate" tip, the actual chance Dana is guilty is **75%, not 90%.** The gap exists because 3 of the 4 suspects started out innocent — some of that innocent majority will always get falsely flagged, and those false flags drag the tip's credibility down. *This is precisely the same math behind medical testing, spam filters, and airport security — a "highly accurate" test on a rare condition still produces plenty of false alarms.*

---

8. Random Variables — 2 min

> **Definition:** a random variable X is a function that assigns a real number to every outcome in the sample space.

Case application: Let X = number of the 4 suspects who are actually guilty of *something* uncovered today (say, 1 vending-machine tamperer plus incidentally 1 other minor rule-breaker found along the way). X can take values 0, 1, 2, 3, or 4 — turning our yes/no detective questions into a number we can average, compare, and eventually attach a full distribution to. "That distribution-building is the next lecture's job — today you built the toolkit that makes it possible."

PART B — PRACTICE PROBLEM SET WITH SOLUTIONS

Problem 1 (Counting). A university committee needs 3 students chosen from a pool of 10 to serve on a review board, where all 3 seats are identical (no distinct roles). How many different committees are possible?

Solution: Order doesn't matter → combination. $C(10,3) = 10! / (3!\cdot 7!) = (10\cdot 9\cdot 8)/(3\cdot 2\cdot 1) = 720/6 =$ **120 committees.**

Problem 2 (Addition Rule). A standard 52-card deck. Draw one card. Let A = "card is a heart" and B = "card is a face card (J, Q, K)." Find $P(A \cup B)$.

Solution: $P(A) = 13/52$, $P(B) = 12/52$, $P(A \cap B) = 3/52$ (the heart face cards: J♥, Q♥, K♥) $P(A \cup B) = 13/52 + 12/52 - 3/52 = 22/52 =$ **$11/26 \approx 0.423$**

Problem 3 (Conditional Probability & Multiplication Rule). An urn has 6 red and 4 blue marbles. Two are drawn *without replacement*. Find P(both blue).

Solution: P(1st blue) = 4/10. Given the first was blue, 3 blue remain out of 9 total → P(2nd blue | 1st blue) = 3/9. P(both blue) = (4/10)·(3/9) = 12/90 = **$2/15 \approx 0.133$**

Problem 4 (Independence). A factory has two machines that each independently produce a defective part with probability 0.05. If one part comes from each machine, find the probability that **at least one** is defective.

Solution: Easier via the complement. P(neither defective) = $(1 - 0.05)(1 - 0.05) = (0.95)(0.95) = 0.9025$ P(at least one defective) = $1 - 0.9025 =$ **$0.0975 \approx 9.75\%$** (Confirms independence is being used correctly: multiplying the two "not defective" probabilities directly, since the machines don't affect each other.)

Problem 5 (Bayes' Formula — the classic version, for contrast with the case). A disease affects 1% of the population. A test for it is 95% accurate in both directions (95% true positive rate, 95% true negative rate — so a 5% false positive rate). A random person tests positive. Find P(disease | positive).

Solution: Let D = has disease, $P(D) = 0.01$, $P(D') = 0.99$. $P(+|D) = 0.95$, $P(+|D') = 0.05$ (false positive rate)

$P(+) = P(+|D)P(D) + P(+|D')P(D') = (0.95)(0.01) + (0.05)(0.99) = 0.0095 + 0.0495 = 0.059$

$P(D \mid +) = P(+|D)P(D) / P(+) = 0.0095 / 0.059 \approx$ **0.161, or about 16.1%**

Discussion point: most students guess ~95%. The true answer is under 20% — because the disease is rare, the large healthy population still produces enough false positives to swamp the true positives. This is the same structural lesson as the Dana example in Section 7, applied to medicine.

Problem 6 (Random Variable, quick capstone). Let X = number of heads in 3 independent fair coin flips. List the sample space, the possible values of X, and P(X = 2).

Solution: S = {HHH, HHT, HTH, THH, HTT, THT, TTH, TTT} — 8 equally likely outcomes (multiplication principle: 2×2×2). X can take values 0, 1, 2, 3. Outcomes with exactly 2 heads: HHT, HTH, THH → 3 outcomes. **P(X = 2) = 3/8 = 0.375**

Problem 7 (Bayes' Theorem — Medical Test, basic). A disease affects 1% of the population. A test for it is 95% accurate for people who have the disease (true positive rate) and 90% accurate for people who don't (true negative rate). If someone tests positive, what's the probability they actually have the disease?

Solution: Let D = has disease, T = tests positive. $P(D) = 0.01$, $P(D') = 0.99$ $P(T \mid D) = 0.95$ $P(T \mid D') = 1 - 0.90 = 0.10$

$P(D \mid T) = P(T|D)\cdot P(D) / [P(T|D)\cdot P(D) + P(T|D')\cdot P(D')]$ $P(D \mid T) = (0.95 \times 0.01) / (0.95 \times 0.01 + 0.10 \times 0.99)$ $P(D \mid T) = 0.0095 / 0.1085 \approx$ **0.088, or about 8.8%**

Discussion point: only about 8.8% — even with a positive result, most people still don't have the disease. This is the classic "base rate" surprise, and a slightly harsher version of Problem 5 above (here the test is worse at ruling healthy people out, which drags the answer down further).

Problem 8 (Bayes' Theorem — Two Factories, intermediate). Factory A makes 60% of a company's phones, Factory B makes 40%. 2% of Factory A's phones are defective, 5% of Factory B's are defective. A random phone is defective — what's the probability it came from Factory B?

Solution: $P(A) = 0.6$, $P(B) = 0.4$ $P(Def \mid A) = 0.02$, $P(Def \mid B) = 0.05$

$P(Def) = 0.6(0.02) + 0.4(0.05) = 0.012 + 0.02 =$ **0.032** $P(B \mid Def) = (0.05 \times 0.4) / 0.032 = 0.02 / 0.032 =$ **0.625**

62.5% chance the defective phone came from Factory B — even though B makes fewer phones overall, its much higher defect rate outweighs its smaller share. This is a clean example of how a prior (production share) and a likelihood (defect rate) trade off against each other.

Problem 9 (Bayes' Theorem — Card from One of Two Decks, trickier). You have two decks: Deck 1 (normal, 52 cards) and Deck 2 (all 52 cards are aces — a trick deck). You pick a deck at random, draw a card, and it's an ace. What's the probability you picked Deck 2?

Solution: P(Deck1) = P(Deck2) = 0.5 P(Ace | Deck1) = 4/52 = 1/13 P(Ace | Deck2) = 1

P(Ace) = 0.5(1/13) + 0.5(1) = 0.5(1/13 + 1) = 0.5 × 14/13 = **7/13** P(Deck2 | Ace) = (0.5 × 1) / (7/13) = (0.5 × 13) / 7 = 6.5/7 ≈ **0.929**

≈92.9% — drawing an ace strongly suggests the trick deck. This is the sharpest of the three: an extreme likelihood ratio (impossible vs. rare) can overwhelm an even 50/50 prior almost completely, which is a good contrast against Problem 8, where a moderate likelihood ratio only partially overturned the prior.
